DFAN: Distributed Fractal Awareness Network — A Biomimetic Control Architecture for Intelligent Energy Transfer Systems

The Conceptual Foundation That Three Data-Driven Papers Missed

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Abstract

Three prior publications documented what Quantum Swarm Heating (QSH) does — fleet simulation results across 348,170 runs (Hunt, 2026a), the governance framework that enabled its construction (Hunt, 2026b), and the domain-expert collaboration model that crossed the paradigm boundary from industrial control to application software (Hunt, 2026c). All three papers delivered evidence. None of them explained the design philosophy.

This paper corrects that omission. QSH was conceived from the outset as a Distributed Fractal Awareness Network (DFAN) — a biomimetic control architecture that treats any energy transfer system as a living organism. In the current residential heating deployment: the heat pump is the heart, flow and return pipes are arteries and veins, per-zone thermal observation is a distributed nervous system, the building envelope is skin, heat loss is perspiration, equilibrium mode is homeostasis, and system identification is the organism learning its own body. The seasonal operating modes — winter, shoulder, summer — are metabolic states, each with distinct energy regulation strategies analogous to biological activity levels. But DFAN is not a heating controller. It is an energy transfer control architecture whose first deployment happens to be residential heating.

This is not a post-hoc metaphor applied to a conventional controller. The biological framing informed the architecture from initial design: the driver-agnostic pipeline exists because organisms are not their environment; the system identification module exists because organisms must learn their own physical parameters; the fractal self-similarity — where the same awareness-and-adapt pattern repeats at room, zone, whole-home, and fleet scales — exists because biological systems scale through structural repetition, not centralised expansion.

The paper maps each DFAN component to its biological analogue and its concrete implementation in the QSH codebase. It positions the three prior publications as validation evidence and identifies the conceptual gap they left: they proved the system

works without explaining why a biomimetic architecture was chosen or why it produces emergent behaviours — shoulder mode discovery, passive equilibrium, demand-gated cycling — that conventional control approaches do not exhibit.

A novelty assessment conducted in April 2026 found, to the author's knowledge, no prior published work combining distributed sensing, fractal self-similarity across scales, and biological awareness-homeostasis into an active adaptive controller for energy transfer systems. The individual elements exist in separate literatures — biomimetic ventilation in architecture, fractal geometry in heat exchangers, swarm optimisation in computational intelligence, multi-agent building control in HVAC research — but the integrated DFAN concept as a named, implemented, and validated control architecture for energy transfer appears to be without precedent.

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1. Introduction

Energy transfer control — the problem of matching a source's output to distributed demand across multiple zones — appears in residential heating, district energy networks, industrial process thermal management, data centre cooling, agricultural climate control, and cold-chain logistics. The physics varies by domain, but the control problem is structurally identical: a source produces energy at a rate and efficiency governed by operating conditions; zones consume energy at rates governed by losses, loads, and environmental disturbances; a controller must match supply to demand.

The dominant approaches to this matching problem — weather compensation curves, PID loops, model predictive control (MPC), and reinforcement learning (RL) — treat it as an optimisation problem. Given inputs (sensor readings, environmental conditions, demand signals), compute optimal outputs (source modulation, distribution, zone setpoints). The building (or process, or cold store) is a plant. The source is an actuator. The controller minimises a cost function.

QSH takes a different view. It treats the source-plus-distribution-plus-zones as a living system. This is not a cosmetic rebranding. The biological framing produces a structurally different architecture from any of the conventional approaches, and that architecture exhibits emergent behaviours — shoulder mode discovery, passive equilibrium, demand-gated cycling — that emerged from the structure rather than being explicitly programmed. The current deployment is residential heat pump optimisation across three UK installations, but the architecture is domain-agnostic: the pipeline manages energy flows between sources and zones based on learned physical parameters and configurable control strategies, without knowledge of the specific domain it governs.

The three prior publications in this series focused on measurable outcomes: energy savings, fleet simulation statistics, governance audit trails, evidence of crossing the industrial-to-application software boundary. They were deliberately data-driven, presenting what the system does and how it was built without requiring the reader to accept the biological framing as a prerequisite. This was a conscious editorial decision — establish credibility through evidence before presenting the conceptual foundation.

This paper now presents that foundation. It is not a retrospective justification. The

DFAN concept predates the first line of code. The biological analogies drove architectural decisions that would have been made differently — or not at all — under a conventional control philosophy. The paper's purpose is to make visible what the prior publications left implicit, and to establish priority for a control architecture that, to the author's knowledge, has not been previously documented.

1.1 Contributions

This paper makes four contributions:

1. It formally defines the DFAN (Distributed Fractal Awareness Network) paradigm as a biomimetic control architecture for energy transfer systems, with explicit component-to-analogue mapping demonstrated through a residential heating deployment.
2. It demonstrates that the biological framing is generative, not decorative — that architectural decisions in QSH trace to biological principles and produce emergent behaviours that conventional control approaches do not exhibit.
3. It identifies the conceptual gap in the three prior publications and positions DFAN as the foundational design philosophy that unifies the technical evidence (Hunt, 2026a), the governance methodology (Hunt, 2026b), and the collaboration model (Hunt, 2026c).
4. It establishes a novelty claim for DFAN as a named, implemented, and field-validated biomimetic control architecture for energy transfer systems.

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2. The Biological Foundation

2.1 Why Biology, Not Control Theory

A conventional control engineer asked to design a multi-zone energy transfer system would begin with a plant model, define transfer functions, design a controller (PID, MPC, or RL), and tune it against a cost function. The result would be correct, stable, and optimisable. It would also be brittle in ways that only become apparent in production: unable to adapt when a zone is added or removed, unable to learn that a particular zone loses energy faster than its specification suggests, unable to discover that running the source at minimum output during low-demand conditions wastes more energy in distribution losses than shutting it down and coasting.

Biological systems face the same class of problem — regulate internal conditions despite variable external disturbances, limited energy budgets, and incomplete knowledge of the environment — and solve it through a fundamentally different architecture. The key properties are:

Distributed sensing and local response. A biological organism does not route all sensory information to a central processor for optimisation. Local sensors (thermoreceptors, baroreceptors, chemoreceptors) trigger local responses (vasodilation, vasoconstriction, metabolic adjustment) while simultaneously informing higher-level regulation. The local response is immediate and adequate; the global coordination refines it.

Self-similar structure across scales. The circulatory system exhibits the same branching, flow-regulation, return-collection pattern from the aorta to the capillary bed. The nervous system exhibits the same sense-process-respond pattern from spinal reflexes to cortical decision-making. The architecture does not change with scale — it repeats.

Continuous self-identification. A biological organism continuously monitors its own state — metabolic rate, core temperature, energy reserves, tissue condition — and adjusts its behaviour accordingly. It does not operate from a fixed model of itself; it learns its current parameters and adapts. Illness, growth, injury, and ageing change the organism's characteristics, and the regulatory systems track those changes.

Metabolic states. A biological organism does not operate at one intensity level. It shifts between metabolic states — active, resting, torpid, hibernating — each with distinct energy regulation strategies. The transitions are triggered by environmental conditions and internal state, and each state has its own homeostatic targets.

These properties are not metaphors for what a controller should do. They are design principles that, when applied to a heating system, produce an architecture that conventional control theory does not suggest.

2.2 The DFAN Acronym

Distributed: Sensing, learning, and decision-making occur at every zone independently. Each room has its own thermal state estimation, its own system identification observations, its own setback calculation. The pipeline aggregates these distributed assessments into whole-home decisions (total demand, flow temperature, heat source scheduling), but the per-zone awareness is primary.

Fractal: The same awareness-and-adapt pattern repeats at every scale. A single room monitors its temperature, estimates its heat loss, and requests energy. A zone (group of rooms) aggregates demand and balances distribution. The whole home coordinates source scheduling, seasonal mode transitions, and equilibrium strategies. A fleet of homes (future scope) would aggregate weather zone patterns and archetype performance. The pattern — sense, estimate, request, coordinate — is structurally identical at each level.

Awareness: The system does not merely measure; it understands. System identification learns per-room physical parameters (heat loss coefficient, thermal mass, solar responsiveness) from passive observation. The pipeline uses these learned parameters to make decisions that a measurement-only controller cannot: predicting heat-up times, calculating equilibrium flow temperatures, determining when demand is too low to justify running the heat pump. This is not a thermostat reading a number; it is an organism that knows its own body.

Network: All components communicate through a shared signal bus (the `CycleContext` in implementation). Sensors inform estimators, estimators inform controllers, controllers inform the driver, and the driver's actions produce new sensor readings.

The circularity is intentional — it is the same negative-feedback architecture that biological homeostasis relies on.

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3. Component Mapping: Biology to Architecture to Code

This section maps each biological analogue to its DFAN architectural concept and its concrete implementation in QSH. The subsections use the residential heating deployment as the illustrative domain, but the architectural concepts are domain-agnostic. The following substitution table applies throughout:

Residential term	General DFAN term	Other domain examples
Heat pump	Energy source	Boiler, chiller, CRAC unit, central plant
Room	Zone	Building, rack, cold-store compartment
Building envelope	System boundary	Container wall, data hall shell, process jacket
TRV / valve	Distribution actuator	Zone valve, damper, heat exchanger bypass
Outdoor temperature	Environmental disturbance	Ambient load, inlet water temp, solar irradiance

Each mapping below represents a design decision that was made because of the biological framing, not despite it. The summary table at the end of this section (Table 2) provides a compact reference.

3.1 The Heart: Energy Source as Central Circulation

In a biological organism, the heart is the central pump that drives circulation. It does not decide where blood goes — that is determined by local vascular resistance (vasodilation/vasoconstriction). The heart provides pressure and flow; the peripheral system determines distribution.

In DFAN, the energy source plays the same role. It produces output at a rate and intensity determined by the flow controller and cascade controller. It does not decide which zones receive energy — that is determined by per-zone distribution controls based on local demand. The source provides the equivalent of cardiac output; the zone distribution controls provide the equivalent of vascular resistance.

The heart has metabolic states. At rest, cardiac output drops to basal levels. Under exertion, it rises. Under extreme cold stress, the body prioritises core temperature over peripheral perfusion. QSH's seasonal operating modes mirror this directly: winter mode maintains full output with equilibrium regulation; shoulder mode reduces to demand-gated intermittent operation; summer mode enters near-dormancy with graduated monitoring intervals. The antifrost override — where the heat pump cannot shut down because external conditions are too severe — is the thermal equivalent of shivering: involuntary metabolic activity to prevent core temperature collapse.

Implementation: `FlowController` (flow temperature), `CascadeController` (PI setpoint tracking), `HardwareController` (on/off execution), `AntifrostOverrideController` (cold-stress involuntary activation). Heart state visualised in the Live View PoC as a pulsing sphere whose colour, rate, and amplitude encode the current metabolic state.

3.2 Arteries and Veins: Flow and Return Circuits

The circulatory system has two halves: arteries carry oxygenated blood outward; veins return deoxygenated blood. Flow rate through each tissue bed is regulated locally by vascular smooth muscle responding to metabolic demand.

In QSH, the flow circuit carries heated water from source to zone emitters; the return circuit carries cooled water back. The flow temperature (arterial pressure analogue) is set by the `FlowController` based on aggregate demand. Per-zone flow rate (local perfusion analogue) is set by TRV valve positions, which the `ValveController` adjusts based on each zone's thermal deficit. The delta between flow and return temperature — the thermal arteriovenous difference — indicates how much energy the system extracted from the circulating medium in one pass.

Implementation: `ValveController` (TRV setpoints per zone), `FlowController` (flow temperature target), `HydraulicController` (minimum flow protection — the thermal equivalent of maintaining minimum cardiac output to prevent circulatory collapse). Visualised as orange (flow) and blue (return) particle streams in the Live View PoC.

3.3 The Nervous System: Distributed Control Signals

A biological nervous system operates at multiple speeds: fast reflexes (spinal cord, milliseconds), intermediate regulation (brainstem autonomic, seconds), and slow adaptation (cortical learning, hours to years). All three levels observe the same sensory inputs but respond at different timescales and with different authority.

QSH's control pipeline mirrors this layered response:

Fast reflex (every 30-second cycle): The deterministic controller chain reads sensors, computes demand, sets flow temperature, adjusts valve positions. This is the spinal reflex — immediate, reliable, adequate for routine conditions.

Intermediate regulation (minutes to hours): The `ShoulderController` tracks demand dissipation over multi-cycle windows to decide whether the heat pump should shut down during mild weather. The `SummerController` monitors sustained low-demand conditions before entering seasonal shutdown. These are brainstem-level decisions — not instant reflexes, but accumulated assessments that trigger state transitions.

Slow adaptation (hours to weeks): System identification learns per-room thermal parameters from passive observation. U-values (heat loss coefficients) converge over

10–50 observations per room. C-values (thermal mass) converge over extended passive cooling windows analysed by Newton's law of cooling fits. Solar gain factors are learned from sunny HP-off periods once U is mature. This is cortical learning — the organism building an increasingly accurate model of its own body.

The neural signals are always active, even in shadow mode (where the system observes but does not control). This is the biological equivalent of maintaining sensory awareness during sleep — the nervous system never fully shuts down, because awareness is the precondition for rapid response if conditions change.

Implementation: Pipeline controller chain (reflex), `ShoulderController`/`SummerController` (intermediate regulation), `sysid.py` (slow adaptation). Visualised as cyan dashed-line neural pulses in the Live View PoC, with increased frequency during shadow/monitoring states to indicate heightened observation without motor output.

3.4 Skin and Heat Loss: The Building Envelope

Biological skin serves two functions: containment (preventing heat loss) and regulation (sweating, vasodilation to manage thermal balance). Heat loss through skin is passive and continuous — the organism cannot prevent it, only minimise it through insulation (clothing, subcutaneous fat, vasoconstriction) or compensate for it through increased metabolic output.

The system boundary is the skin. Energy dissipates across it continuously — the organism cannot prevent the loss, only minimise it through insulation or compensate through increased source output. In the residential deployment, heat flows outward through walls, windows, roof, and floor at a rate proportional to the temperature difference between inside and outside (Newton's law of cooling).

QSH's system identification module learns the effective U-value (heat loss coefficient, kW/°C) of each room by observing temperature decay during HP-off periods. This is the organism learning the thermal conductivity of its own skin. The learned U-values are used throughout the control pipeline: demand calculation, equilibrium flow temperature, setback depth optimisation, and digital twin calibration.

Implementation: `sysid.py` U-value learning (per-room, from HP-off cooling observations), `ThermalController` demand calculation using learned U, `SetbackCalculator` energy-balance sweep using learned U. Visualised as amber "wall leak" particles radiating outward from room nodes in the Live View PoC, with drift rate scaled by learned U-value.

3.5 Homeostasis: Equilibrium Mode

Biological homeostasis is the maintenance of internal conditions within a viable range despite external disturbances. The key insight is that homeostasis is not a target — it is a dynamic equilibrium maintained by continuous negative feedback. Body temperature is not regulated to exactly 37.0°C; it oscillates within a narrow band as metabolic output and heat loss fluctuate around balance.

QSH's equilibrium mode is the direct thermal analogue. In winter, when outdoor temperatures drop below the antifrost threshold (default 7°C), the heat pump cannot shut down — doing so would cause the manufacturer's native antifrost system to seize control at an unmodulated flow temperature, wasting energy. Instead, QSH holds the heat pump on and shifts the control objective from "recover deficit then coast" to "maintain equilibrium at minimum energy." The `FlowController` targets a flow temperature where heat delivery to the building roughly matches heat loss — the deterministic layer calculates this from learned U-values and current delta-T to external. The `CascadeController` switches its PI setpoint from a demand target to an `actual_loss` target.

This is homeostasis: the system is not trying to heat the building to a target. It is trying to hold the building in thermal equilibrium by matching supply to loss, at the lowest energy cost. The heart beats slowly (low amplitude, low frequency in the Live View), the arteries carry just enough flow to compensate for skin loss, and the nervous system monitors for any disturbance that would require the metabolic rate to increase.

Implementation: `AntifrostOverrideController` (detects cold-stress conditions), `FlowController` (equilibrium flow calculation from `ThermalState.actual_loss`), `CascadeController` (PI setpoint switch to loss-matching), `ShoulderController` (off-decisions suppressed during antifrost). The WC floor is capped to an equilibrium ceiling to prevent overshoot.

3.6 Metabolic States: Seasonal Operating Modes

Biological organisms shift between metabolic states in response to environmental conditions. A mammal's metabolic rate varies from basal (sleeping) to maximal (sprinting), with distinct hormonal, cardiovascular, and respiratory profiles at each level. Some species enter torpor or hibernation — dramatically reduced metabolic states that conserve energy when environmental conditions make normal activity unsustainable.

QSH's three seasonal modes map directly:

Winter mode (high metabolic state): The heat pump runs continuously. The control objective is equilibrium maintenance at minimum energy. All controllers are active. The system is fully alert, fully perfusing, compensating for high heat loss rates. This is the organism at sustained moderate exertion in a cold environment.

Shoulder mode (variable metabolic state): The heat pump cycles — running when demand justifies it, shutting down when dissipation tracking shows the building can coast. This is the organism alternating between activity and rest, conserving energy during mild conditions. The `ShoulderController` tracks dissipation failure (demand not reducing despite heating) and triggers mid-run shutdown if the building is not responding — the thermal equivalent of recognising that exertion is not producing the expected metabolic result.

Summer mode (torpor): The heat pump shuts down entirely. Monitoring intervals graduate from 5 minutes to 10 to 30 as sustained low-demand conditions persist. The nervous system remains active (cycle observations continue, system identification can still learn from passive cooling), but motor output ceases. Exit triggers — demand rise, OAT drop, forecast cold snap — function as arousal stimuli from torpor.

The transitions between modes are OAT-gated with hysteresis to prevent chatter — the thermal equivalent of the hypothalamus preventing rapid oscillation between metabolic states. The 1.5°C hysteresis on the antifrost threshold ensures the system commits to a mode transition rather than flickering.

Implementation: `AntifrostOverrideController` (winter entry/exit), `ShoulderController` (shoulder demand-gating, dissipation tracking, mid-run shutdown), `SummerController` (summer entry/exit state machine, graduated monitoring, arousal triggers).

3.7 Self-Identification: The Organism Learning Its Body

A developing organism does not receive a specification document listing its physical parameters. It discovers them through interaction with the environment. Proprioception, interoception, and motor learning build an internal model of the body's capabilities and limitations — a model that updates continuously as the body changes through growth, injury, ageing, or environmental adaptation.

DFAN's system identification module is the direct equivalent. The system starts with prior estimates of each zone's physical parameters (from specifications, archetypes, or commissioning data) and learns the actual parameters from passive observation.

In the residential deployment, QSH's implementation (`sysid.py`) learns per-room thermal parameters:

U (heat loss coefficient): Learned from HP-off cooling periods. The room temperature decays toward outdoor temperature at a rate governed by U/C. With C estimated, U can be extracted. This is the organism learning how fast it loses heat through its skin — not from a specification, but from experiencing cold.

C (thermal mass): Learned primarily from extended passive cooling windows via Newton's law of cooling fits. The time constant tau = C/U is measured directly from the temperature decay curve shape (gated on $R^2 > 0.8$ for statistical confidence). C is then derived as tau × U_effective. This is the organism learning its own thermal inertia — how long it takes to cool down, which depends on both insulation and mass.

Solar gain factor: Learned from sunny HP-off periods with mature U estimates. The room temperature rises despite no heating input; the excess warming rate, after subtracting the expected cooling rate from learned U, reveals the solar gain contribution. This is the organism learning that it warms up in sunlight — a gain term that no thermostat measures.

The learning is continuous, confidence-weighted, and self-correcting. Early observations pull hard (high EMA alpha); as observations accumulate, the learning rate decays to a floor, making the model resistant to noise while remaining responsive to genuine changes (a window replaced, insulation added, a door seal failing). The gate statistics — how many observations were accepted, rejected, and why — are the equivalent of a diagnostic log for the organism's self-knowledge.

Implementation: `sysid.py` (U, C, solar learning with EMA convergence, passive cooling analyser, observation gates), `twin/calibration.py` (bridging learned parameters to digital twin profiles), `sysid_state.json` (persistent learned state — months of accumulated observations that must never be reset without explicit instruction).

Table 2: Component Mapping Summary

Biological System	DFAN Concept	QSH Implementation	Residential Example
Heart	Central source, output modulated by demand	`FlowController`, `CascadeController`, `HardwareController`	Heat pump: flow temp + on/off
Arteries / Veins	Distribution circuit (supply and return)	`ValveController`, `HydraulicController`	Flow/return pipes, TRV positions
Nervous system (reflex)	Fast deterministic control loop	Pipeline controller chain (30s cycle)	Demand calc → flow temp → valve setpoints
Nervous system (autonomic)	Intermediate regulation	`ShoulderController`, `SummerController`	Demand-gated cycling, seasonal shutdown
Nervous system (cortical)	Slow adaptive learning	`sysid.py` (EMA convergence, passive cooling)	U, C, solar gain per room
Skin	System boundary (energy loss surface)	`ThermalController` demand calc, sysid U-learning	Building envelope, fabric U-values
Homeostasis	Equilibrium mode (match supply to loss)	`FlowController` equilibrium flow, `CascadeController` loss-target	Winter low-energy hold
Metabolic states	Seasonal operating modes	`AntifrostOverrideController`, `ShoulderController`, `SummerController`	Winter / Shoulder / Summer
Proprioception	Continuous self-identification	`sysid.py` persistent learning, `sysid_state.json`	Per-room U, C, solar learned from observation
Sensory awareness in sleep	Shadow mode observation	`ShadowController`, sysid active in monitoring-only	System learns without controlling

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4. The Fractal Property

4.1 Self-Similarity Across Scales

The DFAN acronym includes "Fractal" for a specific reason. The awareness-and-adapt pattern in QSH is structurally self-similar across at least three scales, with a fourth projected:

Zone scale (individual demand point): Each zone independently monitors its state, compares to its setpoint, estimates its deficit using learned physical parameters, and requests a proportional share of source capacity via its distribution actuator. The zone is a self-contained entity with its own sensors, its own model, and its own actuator. In the residential deployment, this is a room with a temperature sensor, learned U and C values, and a TRV valve position. In a district network, it would be a building with a heat meter and a zone valve. In a data centre, a rack with temperature sensors and an airflow damper.

Group scale (adjacent demand points): Groups of zones aggregate their individual demands. The system considers inter-zone relationships — a low-demand zone adjacent to a high-demand zone receives adjusted treatment because the high-demand neighbour would pay for the differential through the shared boundary. This is group-level awareness: the behaviour of each zone is modulated by awareness of its neighbours. In the residential deployment, the `SetbackCalculator` considers room adjacency — an unoccupied room adjacent to an occupied room receives a shallower setback because the occupied neighbour would pay for the deep setback through the shared wall.

System scale (full installation): The pipeline aggregates all zone demands into a total system demand, selects an operating mode, determines whether the source should run or coast, sets the source output intensity, and coordinates competing demands (e.g., hot water priority against space heating in the residential case). The system-level controller exhibits the same sense-estimate-decide-act pattern as an individual zone, but at a higher level of abstraction.

Fleet scale (projected): Multiple installations in the same environmental zone, with the same or different archetypes, could share learned parameters to accelerate system identification for new installations, identify anomalous systems that deviate from their archetype, and provide population-level evidence for control strategy tuning. This is the biological equivalent of species-level adaptation — individual organisms sharing information to improve collective fitness.

Figure 1: Fractal Self-Similarity Across Scales

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┌─────────────────────────────────────────────────────────────┐
│  FLEET                                                      │
│  sense: weather zone patterns, archetype population stats   │
│  adapt: cross-installation parameter sharing, anomaly detection │
│                                                             │
│  ┌───────────────────────────────────────────────────────┐  │
│  │  SYSTEM (single installation)                         │  │
│  │  sense: aggregate demand, source capacity, operating mode │  │
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│  │  adapt: seasonal mode selection, source scheduling      │  │
│  │                                                         │  │
│  │  ┌───────────────────────────────────────────────────┐  │  │
│  │  │  GROUP (adjacent zones)                           │  │  │
│  │  │  sense: neighbour occupancy, boundary gradients   │  │  │
│  │  │  adapt: adjacency-aware setback depth             │  │  │
│  │  │                                                   │  │  │
│  │  │  ┌─────────────────────────────────────────────┐  │  │  │
│  │  │  │  ZONE (individual demand point)             │  │  │  │
│  │  │  │  sense: temperature, setpoint, valve position│  │  │  │
│  │  │  │  adapt: learned U, C, solar; demand request │  │  │  │
│  │  │  └─────────────────────────────────────────────┘  │  │  │
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Same pattern at every scale: sense → estimate → request → coordinate
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The architectural consequence of this fractal property is that QSH did not need to be designed from the top down. The room-level pattern was designed first. The zone-level and whole-home patterns emerged by applying the same pattern at higher aggregation levels. The code structure reflects this: per-room calculations are the atoms; zone and home calculations compose them. Adding a room does not require redesigning the system — the new room instantiates the same awareness-and-adapt pattern, and the higher-level aggregations automatically include it.

4.2 Why Fractal Matters for Scalability

Conventional multi-zone controllers scale poorly. A PID-per-zone approach requires N independent tuning exercises with no inter-zone awareness. An MPC approach requires a coupled state-space model whose dimensionality grows with zone count, making real-time computation increasingly expensive. An RL approach requires a state-action space that grows combinatorially.

The fractal approach scales linearly. Each new zone adds one instance of the zone-scale pattern. The group-scale and system-scale aggregations recompute over the updated set. No retuning, no model reformulation, no expanded state space. This is how biological systems scale — a larger organism has more cells following the same differentiation patterns, not a fundamentally different architecture. In the residential deployment, this means a 13-zone house and a 3-zone flat use the same pipeline with different zone counts. The same property would apply to a 50-building district network or a 200-rack data centre.

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5. Emergent Behaviours

5.1 Shoulder Mode Discovery

The most striking emergent behaviour in QSH was the discovery of shoulder mode through fleet simulation. The system was not designed to shut down during mild weather. The original architecture had two modes: winter (heat pump runs) and summer (heat pump off). The fleet simulation across 348,170 parameter combinations revealed a third operating regime: conditions where the heat pump could not usefully run because per-zone demand was below the minimum useful output, but outdoor conditions did not warrant seasonal shutdown.

This was not programmed. It emerged from the interaction between the demand estimation (nervous system), the heat loss model (skin awareness), and the source scheduling logic (metabolic regulation). The system, having learned each room's thermal parameters and calculated the aggregate demand, determined that running the heat pump would produce more distribution loss than useful heating — and shut down. A conventional controller does not discover this; it follows its programmed logic regardless of whether the result is efficient. QSH's biological architecture made shoulder mode an emergent consequence of distributed awareness and self-identification.

The discovery is documented with full quantitative analysis in Hunt (2026a). The fleet simulation showed a 15–45% energy saving potential from shoulder mode operation across UK climate zones and building archetypes.

5.2 Passive Equilibrium

A second emergent behaviour is passive equilibrium — the tendency of the system to hold building temperatures near setpoint with minimal active control during moderate conditions. This emerges from the interaction between learned thermal mass (C), learned heat loss (U), and the demand-gating logic. When thermal mass is high relative to heat loss rate (a well-insulated, massive building), the system learns that it can heat briefly and coast for extended periods. The "coast" phase is not programmed as such — it is the natural consequence of demand dropping below the minimum useful threshold after a heating burst.

This is biological rest: the organism has raised its core temperature, and its thermal mass carries it through the cooling phase without needing to expend additional energy. The duration of the rest phase is not computed analytically — it emerges from the learned parameters and the continuous demand assessment.

5.3 Demand-Gated Cycling

In shoulder mode, the system cycles between heating and coasting in a pattern that is not periodic (like a simple hysteresis controller) but demand-responsive. The duration of each heating burst depends on how quickly the building's demand accumulates. The duration of each coast depends on how slowly the building's temperature decays (a function of learned C and U). The cycle pattern adapts to the building's thermal character without explicit tuning.

This is analogous to respiratory patterns adapting to metabolic demand — breathing rate increases during exertion and decreases at rest, driven by chemoreceptor feedback, not by a fixed timer.

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6. What the Prior Papers Missed

6.1 The Data-Driven Focus

The three prior publications were deliberately data-driven. Hunt (2026a) presented fleet simulation results, energy savings quantification, and the shoulder mode finding. Hunt (2026b) presented the governance framework with its four-stage pipeline, instruction audit trail, and GAMP-inspired validation principles. Hunt (2026c) presented evidence that a process control specialist could cross the industrial-to-application boundary, the collaboration model, and the binding constraint argument.

Each paper was self-contained and evidence-based. Each could be evaluated on its technical merits without requiring the reader to accept the biological framing. This was intentional — establishing credibility through numbers before presenting philosophy.

The cost of that strategy is that the published record, as of April 2026, presents QSH as an unusually effective conventional controller. A reader of Hunt (2026a) sees a system that saves energy through better weather compensation and demand-gated cycling. A reader of Hunt (2026b) sees a well-governed AI development process. A reader of Hunt (2026c) sees a novel collaboration model. None of them sees the underlying design philosophy that explains why these capabilities exist and why they co-occur in a single system.

6.2 The Conceptual Gap

The gap is not just presentation — it is structural. Without the DFAN framing, several aspects of QSH's architecture appear arbitrary:

Why is system identification continuous rather than a one-time calibration? Because the organism is always learning its own body. Parameters change as the building ages, as furniture moves, as windows are opened or sealed. A one-time calibration is a snapshot; continuous identification is proprioception.

Why does the pipeline have 22 controllers rather than a single optimiser? Because biological regulation is distributed across specialised subsystems (cardiovascular, respiratory, endocrine, nervous) that operate semi-independently and coordinate through shared signals. A monolithic optimiser would be more mathematically elegant but less robust to partial failure and less adaptable to structural change.

Why does the system maintain awareness in shadow mode? Because the nervous system does not shut down when motor output is suppressed. Awareness is the precondition for rapid response. Shadow mode is sleep, not death.

Why do seasonal transitions use hysteresis rather than sharp thresholds? Because biological metabolic state transitions are committed transitions with recovery costs. An organism that oscillates rapidly between active and torpid states wastes energy on the transition itself. The 1.5°C hysteresis is the thermal equivalent of the energy barrier that prevents metabolic flickering.

These design decisions are individually defensible on engineering grounds. But their coherence — the fact that they all point in the same direction — is explained by the biological framing. They are not independent engineering choices; they are manifestations of a single design philosophy.

6.3 The Missing Trunk

The three prior papers are branches of a tree whose trunk was never published. Hunt (2026a) is the evidence branch: "here is what the system achieves." Hunt (2026b) is the process branch: "here is how it was built safely." Hunt (2026c) is the collaboration branch: "here is who built it and why that matters." The trunk — "here is the design philosophy that made all three possible" — is this paper.

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7. Novelty Assessment

7.1 Prior Art Landscape

A systematic search of academic databases, preprint repositories, patent filings, industry publications, and open-source project documentation was conducted in April 2026 to assess whether the integrated DFAN approach — combining distributed sensing, fractal self-similarity across scales, and biological awareness-homeostasis in an active adaptive controller for energy transfer systems — has been previously published.

The search identified parallel work in four adjacent domains:

Biomimetic building design. Architectural research applies biological principles to building form — termite-mound-inspired ventilation (Eastgate Centre, Harare), fractal geometry in facade design for natural ventilation, biophilic design principles for occupant wellbeing. These address building design, not building control. The building is passive; there is no adaptive controller learning the building's thermal character.

Fractal heat transfer. Thermal engineering research applies fractal geometry to heat exchanger and heat sink design, where self-similar branching patterns maximise surface area for heat transfer. This is geometric optimisation of hardware, not a control architecture. The fractal property is in the physical structure, not in the information architecture.

Swarm and multi-agent HVAC control. Computational intelligence research applies swarm optimisation (PSO, ACO) and multi-agent systems to building energy management. These use biological metaphors for optimisation algorithms (ant colony, particle swarm) but do not structure the control system itself as a biological organism. The agents optimise; they do not sense, learn their own parameters, or shift metabolic states.

Model predictive control for buildings. MPC research uses physics-based building models for predictive control. This shares QSH's emphasis on learned models but uses them differently — as prediction horizons for optimisation rather than as self-knowledge that informs awareness and metabolic state selection.

7.2 The Novelty Claim

No prior work was found that combines the following elements into a single, named, implemented, and field-validated control paradigm:

1. Distributed per-zone sensing and local decision-making (not centralised optimisation).
2. Fractal self-similarity where the same awareness-and-adapt pattern operates at zone, group, and system scales.
3. Continuous system self-identification that learns the installation's physical parameters from passive observation (not one-time commissioning).
4. Biological metabolic states (operating modes) with committed transitions and distinct energy regulation strategies.
5. Emergent behaviours (shoulder mode, passive equilibrium) arising from the architecture rather than being explicitly programmed.
6. A named architecture (DFAN) with an explicit component-to-analogue mapping.
7. Production deployment on physical hardware governing real energy transfer processes with quantified operational evidence.

Individual elements exist across the literature. Their integration does not.

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8. The Live View as Communication Layer

A Canvas-based visualisation (the QSH Live View proof of concept) was developed to make the DFAN architecture tangible. It renders the heat pump as a pulsing heart, flow and return as particle-animated arteries and veins, control signals as neural pulses, heat loss as skin-leak particles, and rooms as temperature-responsive nodes in a radial network topology. The full set of operating states from the `ShadowController` two-tier model are visualised with state-specific heart behaviour, particle modulation, and annotation.

The visualisation is an explanatory artefact, not the runtime DFAN engine. It renders data from the production system (via WebSocket) but does not participate in control decisions. Its value is communicative: it makes the biological architecture visible in a way that code listings, architecture diagrams, and data tables cannot. When the heart slows during equilibrium, when particles stop during shoulder coast, when neural pulses brighten during shadow monitoring, the observer understands the system's state through the same intuitive vocabulary that biological regulation uses.

Figure 2 (included in the Zenodo deposit as a supplementary image) shows the Live View rendering a 13-zone installation in winter heating mode. The pulsing heart is visible at centre with COP-coloured corona, orange flow particles move outward along curved Bezier arteries to room nodes, blue return particles trace veins back to centre, cyan neural pulses traverse dashed control paths, and amber wall-leak particles radiate outward from each room at rates proportional to their learned U-values. Room nodes are sized by floor area and coloured by thermal state (blue → orange → green as temperature approaches setpoint).

A production integration of this visualisation into the QSH web interface is in progress (INSTRUCTION-48 in the project's development archive).

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9. Relationship to Existing Publications

This paper is the fourth in a series. The relationship between the four papers is:

This paper (Hunt, 2026d): The design philosophy. Why the architecture exists. What biological principles it embodies. What emergent behaviours it produces.

Hunt (2026a) — Physics-Based Thermal Optimisation: The evidence paper. Fleet simulation across 348,170 runs demonstrating energy savings, weather compensation analysis, and the shoulder mode discovery. This paper provides the quantitative validation for DFAN's emergent behaviours.

Hunt (2026b) — Air-Gapped AI Development Pipeline: The process paper. The GAMP-inspired governance framework that enabled a domain expert to build the DFAN system through structured AI collaboration. This paper explains how the DFAN architecture was implemented safely.

Hunt (2026c) — Domain-Expert AI Collaboration: The boundary-crossing paper. The evidence that a process control specialist could cross from industrial control to application software, and the argument that domain expertise is the binding constraint. This paper explains who built the DFAN system and why that collaboration model works.

Read together, the four papers form a complete account: the philosophy (this paper), the evidence (2026a), the process (2026b), and the people (2026c). The first three were deliberately published before this one — evidence before philosophy — to establish the technical foundation on which the DFAN claims stand.

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10. Limitations

1. **Single deployment domain.** DFAN is presented with evidence from residential heat pump control only. The architecture is domain-agnostic by design — the pipeline manages energy flows between sources and zones without knowledge of the specific domain — but transferability to other energy transfer domains (district heating, industrial process thermal management, data centre cooling, cold-chain logistics, agricultural climate control) is projected from the architecture's structure, not demonstrated through deployment.
2. **Three installations.** The production evidence comes from three UK residential installations (~200,000+ combined cycles). This is sufficient to demonstrate that

the architecture works in practice but insufficient for population-level statistical claims.

3. **Fleet scale not yet implemented.** The fractal property is demonstrated at zone, group, and system scales. The fleet scale (Section 4.1) is projected based on the architectural pattern but not yet implemented or validated.
4. **Biological framing is not biological computation.** QSH is not a neural network, not a genetic algorithm, and not a biomolecular simulation. It uses biological principles as design constraints for a conventional (deterministic + RL) control system. The claim is that the biological framing produces better architecture, not that biological computation is being performed.
5. **Single author, single conceptual framework.** DFAN was conceived by a single individual. Independent evaluation by control theorists, biological systems researchers, and energy systems specialists would strengthen the novelty and validity claims.

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11. Conclusions

This paper presents DFAN — Distributed Fractal Awareness Network — as a formally defined biomimetic control architecture for energy transfer systems. DFAN treats the source-plus-distribution-plus-zones as a living organism, with explicit mappings from biological components (heart, arteries, veins, nervous system, skin, homeostasis, metabolic states, proprioception) to architectural concepts and production code. The mappings are demonstrated through a residential heat pump deployment but apply to any domain where a source distributes energy to multiple zones across a system boundary.

The biological framing is not decorative. It is generative — producing architectural decisions (distributed sensing, continuous self-identification, metabolic state transitions, fractal scalability) that conventional control approaches do not suggest, and emergent behaviours (shoulder mode discovery, passive equilibrium, demand-gated cycling) that were not explicitly programmed but arose from the architecture's structure.

Three prior publications documented what QSH achieves and how it was built without presenting the foundational design philosophy. This paper corrects that omission. It establishes DFAN as the conceptual trunk from which the technical evidence (Hunt, 2026a), the governance methodology (Hunt, 2026b), and the collaboration model (Hunt, 2026c) extend.

A novelty assessment found no prior published work combining distributed sensing, fractal self-similarity, biological awareness-homeostasis, and continuous self-identification into a named, implemented, and field-validated control architecture for energy transfer systems. Individual elements exist across adjacent literatures; their integration does not.

The DFAN concept originated not from academic theory but from practical frustration — an automation engineer applying 28 years of process control intuition to an energy transfer problem that conventional controllers managed poorly. The biological framing was not chosen for novelty; it was chosen because it produced better design decisions than any control theory framework the author had encountered. That the resulting system exhibits emergent behaviours, scales fractally, and learns its own physical parameters is the evidence that the framing was not just useful but correct. That it did so in its first deployment domain — residential heating — while maintaining an architecture that is structurally independent of that domain, suggests DFAN has reach beyond the trial that validated it.

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12. Data Availability

- **QSH source code:** github.com/stuartj1-1981/Quantum-Swarm (Apache 2.0)
- **Live View PoC:** `qsh_live_view_poc.html` in the repository root
- **Fleet simulation database:** Zenodo deposit (Hunt, 2026a), ~2 GB SQLite
- **Governance framework:** Zenodo deposit (Hunt, 2026b)
- **This preprint:** Zenodo deposit, CC BY 4.0. doi:10.5281/zenodo.19443159

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AI Contribution Disclosure

This preprint was developed through human–AI collaboration using the air-gapped pipeline described in Hunt (2026b). AI agents (Anthropic Claude, Opus 4.6 model tier) contributed implementation support including literature search assistance, structural suggestions, and text drafting.

All intellectual content — the DFAN concept, the biological-to-architectural mappings, the novelty claims, the arguments, and all assertions of fact — are the sole responsibility of the named author. No AI agent is listed as author.

Citation verification. Every reference in this version has been resolved to a primary source: each DOI has been resolved and its returned title and authors compared against the citation as written, and each standard retrieved and its title and issuing body confirmed. This check was not performed on version 1. Two entries in version 1 did not survive it and have been corrected — see the revision note below.

The Quantum Swarm Heating system described in this paper was built through the collaboration model documented in Hunt (2026c), using the governance framework described in Hunt (2026b). DFAN as documented here predates and informed the architectural decisions validated in Hunt (2026a). The residential heating deployment is the validation domain, not the scope of the architecture.

Revision note — version 2 (7 August 2026)

This version corrects the reference list. No claim, result, figure or argument has been altered.

Ref	Version 1	Version 2	Reason
5	Pearce, M. et al. (1996). *Eastgate Development, Harare: Biomimicry in Building Design.* Arup Journal, 31(1).	Hawkes & Forster (2003), *Eastgate Centre, Harare, Zimbabwe*, doi:10.4324/9780203362488-18	The version 1 entry could not be verified. The Eastgate Centre and its architect are real, but that article and volume could not be confirmed. Replaced with a verified published source on the same building.
7	Chen, Y. & Shi, Y. (2007). *Particle Swarm Optimization for Multi-Agent HVAC Control.* Applied Energy, 84(11).	Kusiak, Xu & Tang (2011), *Optimization of an HVAC system with a strength multi-objective particle-swarm algorithm*, Energy 36(10), doi:10.1016/j.energy.2011.08.024	**The version 1 entry does not exist.** No such paper is published in Applied Energy 84(11) or elsewhere. Replaced with a verified paper applying particle-swarm optimisation to HVAC control.
17	—	Wang et al. (2022), *A general multi agent-based distributed framework for optimal control of building HVAC systems*, doi:10.1016/j.jobe.2022.104498	Added so the multi-agent HVAC literature discussed under “Swarm and multi-agent HVAC control” rests on a verified source.

Neither corrected entry was cited by number in the body text, so no argument in this paper depended on either. The discussion of biomimetic building design and of swarm and multi-agent HVAC control is unchanged and remains accurate.

Why this happened. The version 1 bibliography was assembled with AI literature-search assistance and was not independently verified before deposit. The failure mode is specific and worth recording: fabricated entries carried *real* journal, volume and article-number coordinates attached to fabricated author and title pairs. Such entries are correctly formatted, internally consistent, and resolve to real containers — they cannot be detected by reading, only by resolving the identifier. The governance regime under which the software described here is built verifies code and did not, at that time, verify citations. It now does.